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What Don't We Know: Identifying Specific Gaps in GenAI Literacy

Proposed Features for Learning Support Tools in AI Systems

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Introduction

As generative artificial intelligence (GenAI) becomes more ubiquitous in the workplace, especially in small and medium-sized enterprises (SMEs), it is vital to ensure that these tools are properly understood by those using them. SMEs employ “between half and two-thirds of all employees... in Austria” and generally work with fewer resources to integrate new technologies successfully and extract the highest-quality results from them (Kopkow et. al., 2025). Kopkow et al. (2025) identified several notable characteristics of how employees are using GenAI. The most concerning finding was recorded in the abbreviated generative AI literacy assessment test (GLAT). This instrument uses a multiple-choice test to gauge participants’ knowledge of GenAI, their ability to use and apply the technology, their ability to evaluate and create with the technology, and their knowledge of ethics (Jin et al., 2025). Kopkow et al. (2025) reported that, in three of the four categories, participants on average answered only approximately one of three questions correctly.

If workers are using GenAI frequently while lacking a clear grasp of the tool’s strengths and weaknesses, there exists a high potential for errors of all kinds, including running afoul of new regulations. As companies seek to integrate GenAI into their workflows, the proper tools and training will be necessary to ensure both that employees use the technology to its fullest effect and that costly mistakes are avoided. Especially for SMEs, which already have limited resources, the potential for consequential errors made with GenAI should not be taken lightly. It is therefore advantageous to understand exactly what most employees do not know about GenAI before it is significantly integrated into a company. That understanding is also key to understanding what kinds of interventions are useful.

Problem Statement

Given the demonstrated gaps in the participants’ GenAI literacy in Kopkow et. al. (2025) the following questions arise:

RQ1- Which specific concepts are most frequently misunderstood, with regard to participants’ GenAI literacy?

RQ2- How can those gaps be addressed in the context of GenAI application design?

By identifying specific gaps in SME employees’ GenAI literacy, we can offer targeted guidance for concise future training. In the short term, these findings could allow for stopgap-style documentation to address the most common issues . In the long term, these findings could help to create GenAI applications that assist the user in building an accurate mental model of the AI’s capabilities.

This paper addresses these questions first through analysis of a dataset containing information about participants’ GenAI literacy, acceptance, and self-reported use and concerns. It then discusses those findings in the context of design considerations and concepts for ways in which GenAI systems can directly help teach GenAI literacy.

Methodology

Description of Data Set

The data set is in two parts. One contains the contents of a survey where participants answered open-ended questions about their use and concerns about GenAI as well as a technology acceptance questionnaire. The other contains the responses and scores from the GLAT. It is not guaranteed that the same participants appear in both datasets. The GLAT was conducted in-person and anonymized. The survey was conducted completely anonymously online.

Methods

Generative AI Literacy Assessment Test

The Generative AI Literacy Assessment Test (GLAT) was administered to those willing to take it after the workshops conducted by Kopkow et. al. (2025). The GLAT is a performance-based questionnaire developed by Jin et. al. (2025), measuring participants' understanding of core concepts about the capabilities and proper use of GenAI tools. It was designed for use among higher education students. The GLAT underwent three validation studies which reported strong content, structural, and external validity, making it a useful tool for objectively measuring GenAI competence. The test comprises twenty questions separated into four dimensions. Twelve questions across all four dimensions were selected for the present study.

Know & Understand

This dimension tests participants' knowledge of GenAI structure and architecture, as well as the general capabilities of a GenAI model. It contains concepts such as "foundation models (e.g., LLM and diffusion model), generation capability, zero-shot learning, prompt-based development, content generation, token, Artificial General Intelligence (AGI), model alignment, [and] Retrieval-Augmented Generation (RAG)." (Jin et. al., 2025)

Use & Apply

This dimension tests participants' understanding of what tasks GenAI models can do and how they do them. It contains concepts such as "contextual understanding in content creation, knowledge updating and integration, information retrieval and synthesis, token management and limitation, [and] multimodal content generation." (Jin et. al., 2025)

Evaluate & Create

This dimension tests participants' knowledge of how to supervise and evaluate the output of a GenAI model in order to create accurate, high quality content. It contains concepts such as

“trustworthiness of LLM outputs, LLM knowledge cutoff, information cross-check (Hallucination), GenAI-generated content authenticity, [and] voice cloning with GenAI.” (Jin et. al., 2025)

Ethics

This dimension tests participants’ knowledge of GenAI limitations and ethical issues that may arise from use of GenAI models. It contains concepts such as, “model biases, black box issue in GenAI, copyright issues in GenAI-generated content, content safety, [and] privacy concerns with GenAI.” (Jin et. al., 2025)

Without the presence of grouping variables, we cannot draw conclusions about which populations are most affected by GenAI illiteracy. Instead, descriptive statistics are herein used to assess the mean score on the entire GLAT and each dimension as well as the variation in scores. A non-parametric Friedman test was used to identify significant differences in mean scores between the subdimensions. No other groups were created or compared due to the small sample size. (Jin et. al., 2025)

Unified Theory of Acceptance and Use of Technology 2

In a separate online survey, participants answered questions regarding their GenAI usage at work. The survey contained an adapted version of the Unified Theory of Acceptance and Use of Technology 2 (UTAUT2) (Venkatesh et. al., 2012). The UTAUT2 is a technology acceptance model describing the constructs and moderators that give rise to behavioral intention to use a technology in a consumer context (Venkatesh et. al., 2012). The six constructs which serve as predictors and behavioral intention are defined below:

Performance Expectancy

This construct describes user expectations of the system’s performance. This includes expectations that the system is useful, increases productivity, and helps the user accomplish their goals. Users who perceive a system as performing well and being useful will be more likely to accept that system. In the context of GenAI, this refers to users’ perception of how well the model can generate their desired results.

Effort Expectancy

Effort expectancy describes user expectations of how much effort it takes to learn and use the system. If a system requires less effort than previous methods to accomplish a task, then it is more likely to be accepted by users. In the context of GenAI, this encompasses users’ perceived ability to become proficient and understand the system.

Social Influence

This construct refers to the social factors that often drive behavior, in this case the perceived opinions of those around the user that could influence their behavior. This includes the opinions

of those the user cares about and those the user believes influence their behavior. In the context of GenAI, this could range widely from general public perception to workplace attitudes to the opinions of friends and family.

Facilitating Conditions

Facilitating conditions refer to the structures and support that enable ease of use. This can include documentation or access to help. In the case of GenAI this refers to the perceived accessibility of information about the system.

Hedonic Motivation

Hedonic motivation refers to the sense-pleasure experiences that keep consumers engaged with a given technology. It comprises aspects such as fun, enjoyability, and whether or not the system is entertaining. This translates exactly to GenAI: do users enjoy using the system?

Habit

This construct refers to users' perceptions about their behavior and specifically whether they have formed a habit of using the system. A system that creates habitual use is more likely to be accepted. This also translates directly to GenAI; do users habitually use the system?

Given the small sample size, full predictive modeling of UTAUT2 was not pursued. Instead descriptive statistics were calculated for each construct, and used to inform the design considerations. These constructs represent users' likelihood of acceptance for a technology most of them already use. As such the UTAUT2 results were given only secondary consideration when creating the design considerations.

Content Analysis of Open-ended Questions

The open-ended survey questions provide a more detailed look at participants' usage and concerns about GenAI, however most responses are quite short, consisting of a bulleted list or just a few words. Given the short nature of the responses, there was not enough data to empirically code which misconceptions from the GLAT are also present in the open-ended responses. I endeavored to avoid making any inferences or using possible subtext in this analysis because the answers are translated from German to English using DeepL, and the filter of my subjective inference *and* machine translation may warp results. Content analysis was bottom-up and iterative, beginning with a topic analysis and then narrowing to identify specific tasks and concerns named by participants. With the possibility of finding more specific misunderstandings, content analysis hopefully opens up more avenues for designing learner-centered GenAI or a training program which helps address GenAI literacy.

Results

GLAT

Across the full GLAT scale, participants answered an average of slightly more than half of the questions correct, with a moderately small standard deviation ($M=6.84$, $SD=2.49$). This indicates there may be gaps in participant knowledge about GenAI. The dimensions of the test were assessed to gain further detail into specific gaps.

Participants had the highest average score and the lowest standard deviation in the Know & Understand dimension ($M=2.14$, $SD=0.857$), suggesting that participants understand the definition of GenAI. Evaluate & Create ($M=1.64$, $SD=1.045$) and Ethics ($M=1.68$, $SD=1.096$) had almost identical means and standard deviations. Participants on average, scored the lowest on the Use & Apply dimension ($M=1.38$, $SD=0.967$). Use & Apply also had the lowest median score ($Mdn=1$). All other dimensions had the same median score ($Mdn=2$).

Dimension	Mean	Median	Standard Deviation
<i>Know & Understand</i>	2.14	2	0.857
<i>Use & Apply</i>	1.38	1	0.967
<i>Evaluate & Create</i>	1.64	2	1.045
<i>Ethics</i>	1.68	2	1.096

Fig 1. GLAT Dimensions & corresponding descriptive statistics for the sum score of the dimension (0-3 points)

A Shapiro-Wilk test resulted in statistically significant ($p<0.001$) non-normal distributions for all dimensions. Analysis of a QQ plot for each dimension showed that, despite being a sum of the points within each dimension, the data behaved too much like an ordinal variable with one of four ranks. Therefore a Friedman rank sum test was chosen. This test found significant differences between the dimension means ($\chi^2(3, N=50) = 18.32$, $p<0.001$). Pairwise comparisons using a Wilcoxon rank sum test found a significant difference only between Know & Understand and Use & Apply ($p=0.001$). Differences between all other dimensions were non-significant.

Item level interpretation must be done with extreme caution. The frequency of correct responses was calculated for each question as well as the frequency of each response to note whether there were any consistent misconceptions. Several questions demonstrated a pattern of less than 50% of respondents answering the question correctly and a plurality choosing the same wrong answer. With regard to the Use & Apply dimension, these frequencies are as follows:

Question	Response	n(%)
When using generative AI to create a marketing pitch, which of the following strategies is LEAST likely to be effective?	Providing the AI with a list of competitors' products.	22(44%)
	Requesting the AI to use persuasive language techniques.	21(42%)
	Asking the AI to include unique selling points and benefits.	6(12%)
	Supplying the AI with information about the target audience.	1(2%)
After deploying a customer service chatbot, you notice that it frequently provides outdated information about company policies. What is the best course of action to address this issue?	Schedule regular updates to the chatbot's training data to include the latest company policies.	28(56%)
	Set up a system where complex or policy-related queries are escalated to human agents for accurate responses.	10(20%)
	Conduct a comprehensive audit of the chatbot's performance metrics to identify areas for improvement.	8(16%)
	Implement a feedback loop where users can flag outdated information for review.	4(8%)
Suppose you have a large dataset of emails and you want to build an application to answer questions based on this dataset. Which of the following scenarios best illustrates the advantage of using RAG over prompting (i.e., without RAG)?	You want to ensure the model can answer questions even if it has never seen similar questions before.	22(44%)
	You need to answer questions that require specific information from different parts of the email dataset.	19(38%)
	You want to reduce the size of the language model to save computational resources.	5(10%)
	You need to generate creative writing pieces based on the email content.	4(8%)

Fig 2. *Use & Apply* dimension questions with possible responses and number(proportion) or respondents. **Correct response highlighted in green with bold text**

UTAUT2

Given the small sample size (N=34), the UTAUT2 was not modeled to see which constructs predicted behavioral intention. Only one respondent said in the open-ended questions that they

do not have use for GenAI. When asked how often they use GenAI, every participant in the sample reported using it at least once a month. The broad majority of the sample responded that they use GenAI multiple times a week (n=13[38.2%]) or multiple times a day (n=14[41.2%]). It is therefore logical to assume a high degree of acceptance of GenAI among this sample. Descriptive statistics instead can help clarify participants' perceptions of a technology they have already accepted. Descriptive statistics for each UTAUT2 construct are shown below:

Construct	Mean	Median	Standard Deviation	Minimum	Maximum
<i>Performance Expectancy</i>	5.245	5.333	0.712	3	6
<i>Effort Expectancy</i>	4.066	4.250	1.173	1.25	6
<i>Social Influence</i>	3.480	3.000	0.996	2	6
<i>Facilitating Conditions</i>	3.699	3.625	1.114	1.5	6
<i>Hedonic Motivation</i>	4.412	4.667	1.303	0	6
<i>Habit</i>	2.559	2.667	1.474	0	6

Fig 3. UTAUT2 constructs & corresponding descriptive statistics for the mean score of the construct (0-6 points).

The relatively high averages (>3) across five of the constructs supports the assumption that acceptance of the technology is high. Notably, Performance Expectancy had a very high mean (M=5.245, Mdn=5.333, SD=0.712). The medians of all constructs are also very close to the corresponding means, and with no standard deviations higher than 1.5, the sample appears remarkably consistent in its acceptance and perceptions of GenAI.

Content Analysis

Inductive content analysis was conducted on two open-ended questions, one addressing participants' named use cases for GenAI, and the other addressing their concerns about the technology.

Participants first answered the question "How do you currently use applications that incorporate GenAI into your day-to-day work?" A total of 89 distinct thematic units were extracted from participants' responses, usually in a list form. One unit contains one use case. These use case units were then classified into groups of related units to establish a more general label. For example, "General questions", "Internet searches", and "Google replacement" would all be classified as "Internet search/ Search engine replacement". After grouping, units in the same group from the same respondent were merged to capture unique responses from each

participant. This resulted in a reduction of only a few responses, and most responses did not require merging. After merging there were 86 unique units.

Results show that “Search engine replacement/ Research/ Questions on general knowledge” (n=17 responses) and “Text generation/ Editing/ Revision” (n=16 responses) were named much more than other use cases. Most other use cases were named four or five times and appear to be somewhat specific to the tasks a given respondent needs to fulfil in their job role. All use cases named only once were classified as miscellaneous.

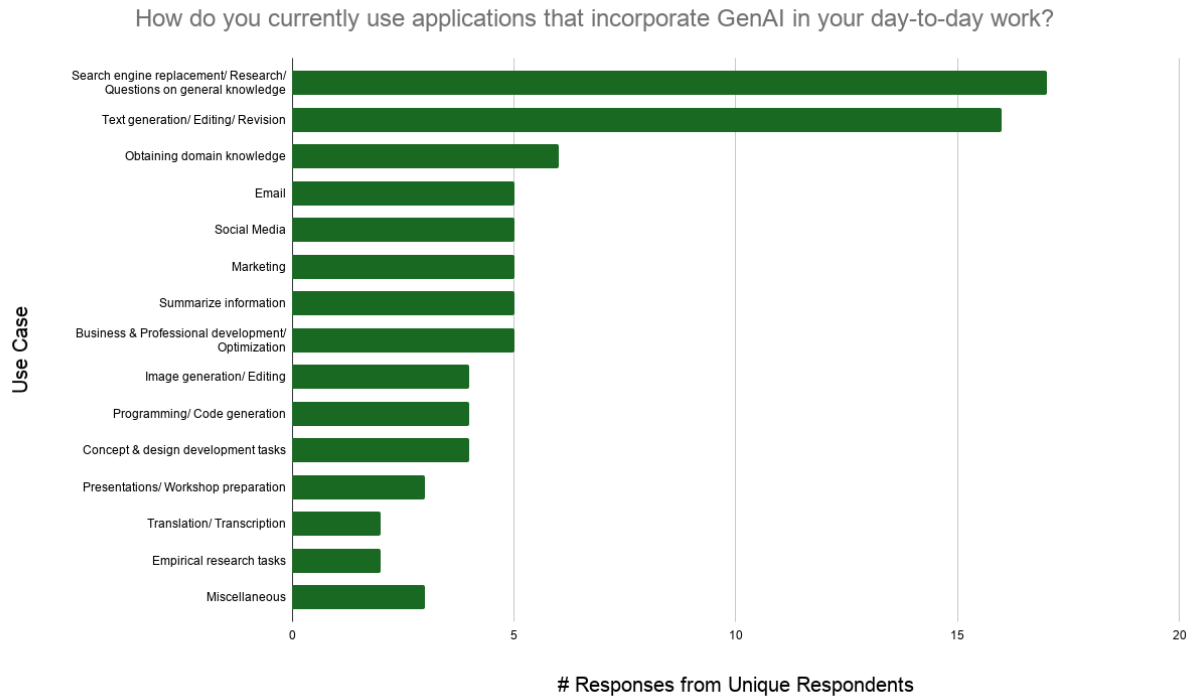


Fig 4. Results of content analysis on GenAI use cases

Participants then answered the question “What concerns or worries do you have in your day-to-day work regarding the use of applications that incorporate GenAI?” 55 distinct thematic units were extracted and classified into groups following a similar procedure to the previous question: first classifying them into a group and then merging any responses from the same respondent in the same group. After merging there were 53 unique responses.

Results show that “Privacy/ Data security and protection/ Use of input data for model training” (n=15 responses) and “Model’s accuracy/ Inaccurate outputs” (n=11 responses) were the two most common concerns. These were followed by “No concerns/ Unspecified very few concerns” (n=10 responses) and “Improper or reckless use of the technology” (n=7 responses). All other concerns had three or fewer responses, and all concerns with only one response were categorized as “Other”.

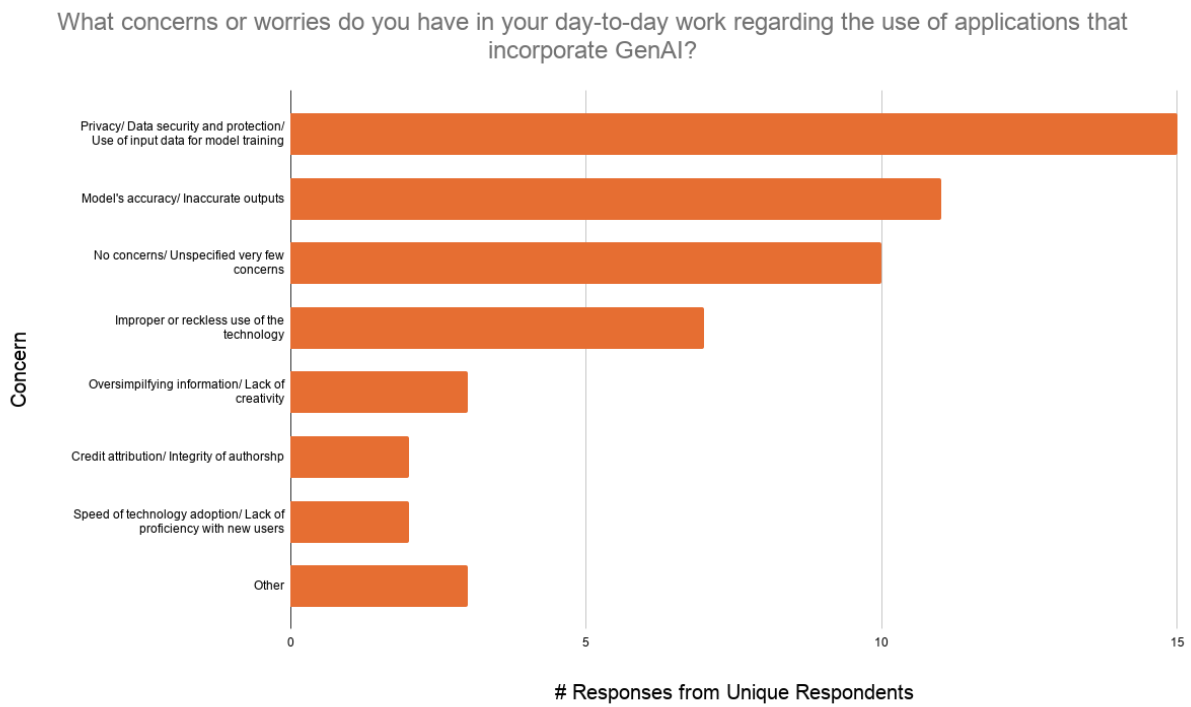


Fig 5. Results of content analysis on concerns about GenAI

Discussion

Gaps in Competency

Results on the GLAT suggest several gaps in GenAI literacy. The mean of the full scale suggests that participants typically only answered about half of the questions correctly. A closer look at each dimension reveals that while participants can reliably define GenAI and understand the structure of a GenAI model, they do not reliably understand how to most effectively use the technology. The significant difference found between the Know & Understand dimension and the Use & Apply dimension suggests that the ability to define GenAI and even name some of its capacities does not necessarily translate to adequate use of the technology. Item-level analysis reveals that on the topics of prompt engineering, mitigating out of date responses, and alternative to prompting, a large proportion of participants (>50% for each item) did not identify the best or most efficient strategy to solve the problem presented.

The GLAT results suggest that participants struggled to identify the strengths and weaknesses of GenAI. Long and Magerko (2020) identify several necessary competencies based on their assessment of GenAI literature, such as the need to understand GenAI's strengths and weaknesses. Identifying which problems GenAI excels at and which problems it struggles with is key to knowing when the use of GenAI is appropriate and when it makes more sense to leverage human skills (Long & Magerko, 2020).

Descriptive analysis of the UTAUT appears to support the assumption that the sample has largely already accepted GenAI. Most of the mean scores for each construct are relatively high, with the lowest still representing an opinion slightly more positive than a true neutral opinion of the technology. The only score below three was that for the habit construct suggesting that, despite an overwhelming proportion of the sample reporting GenAI use at least multiple times a week, participants do not appear to believe that they have formed a habit. While this does not address a specific competency, it does suggest that participants may have a somewhat distorted view of their own GenAI usage with respect to how much they use it.

It would be interesting to put the extremely high performance expectancy score in conversation with the GLAT, namely the somewhat low mean score of Evaluate & Create. However, the disjointed nature of data collection means that it is impossible to verify whether the GLAT participants are the same participants who filled out the online survey. As such, it can only be said that further research should explore users' perceptions of GenAI performance in relation to metrics of both GenAI's actual performance and users' knowledge of how to use the technology most effectively. This would reveal more about specific gaps in GenAI competencies.

Content analysis of what participants use GenAI for revealed two predominant use cases for text generation and editing (n=17) and as what is effectively a replacement to search engines for general knowledge research (n=16). All other use cases had fewer than six responses. This suggests that text generation and general knowledge questions are widespread use cases across many kinds of users within the sample, whereas the other use cases appear to be more

domain-specific. Again, while not indicative of a specific GenAI competency, an understanding the reported uses of the technology is necessary to practically address concerns of competency.

Content analysis of participants' concerns about GenAI reveals a concerning trend among the sample. The two most cited concerns, privacy/data security (n=15) and concerns about GenAI's accuracy (n=11), are valid concerns to raise and demonstrate a proportion of the sample that are considering the potential pitfalls of the technology. However, the third most common response was effectively that they had no concerns. Of the ten respondents who stated they had few or no concerns, only one elaborated on another concern which was counted in another group. Nine out of thirty-four respondents (~26%) state categorically that they have no concerns about the use of GenAI and did not contribute responses to any other group.

While it is impossible to say why these respondents have no concerns, and this content analysis cannot be directly tied to the scores on the GLAT, both of these data points independently suggest that participants may potentially be trusting the systems excessively. To have no concerns betrays a level of confidence that, at this point in GenAI's technological development, is not warranted for any model currently available. Long and Magerko (2020) detail competencies such as understanding GenAI decision-making, understanding the human role in creating GenAI models, machine learning literacy, data literacy, and critical interpretation of data. While it is not possible to judge which specific competencies are lacking, too much trust in the system suggests that users do not have an adequate grasp of GenAI's fallibility and why these systems are fallible.

Proposed Features

Features should be implemented directly into systems to teach users the necessary competencies to adequately use GenAI. These features should provide information and structure to help users understand model outputs and calibrate trust (National Academies of Sciences, Engineering, and Medicine 2022).

Informative Add-ons

These proposed features could be designed and implemented for any GenAI model currently available and added to the existing interface as informational supplements that would foster increased GenAI literacy.

Output Confidence Level

If users struggle to identify the strengths and weaknesses of GenAI, then they would benefit from guidance about the system's performance on the task they have prompted. Most consumer GenAI models now have a so-called thinking feature, which provides transparency about the models' decisions. Another layer to that transparency would be confidence levels about the accuracy of the output at each stage of thinking. The system could directly communicate to the user the likelihood of good performance via a user-friendly confidence ranking system, for

example tags on the output responses which flag them from not confident to very confident depending on the performance metrics of a given model.

Bias Warnings

Long and Magerko (2020) identify ethics as a necessary competency. This may also be undermined by excessive trust in the system. Tasks considered high risk, or that are historically known to have biases embedded through historical practice into training datasets should have warnings, for example hiring processes. Guardrails already exist where systems already refuse to execute prompts based on filters for acceptable behavior established by the proprietors of the model. Similar warnings and questions about whether or not to proceed could be implemented for tasks where known biases exist. Moreover, tags similar to the ones described above, but describing the risk of bias in the dataset, could be implemented as a static warning even if a user decides to execute a potentially risky prompt.

Usage Tracker

While not a specific competency, the disconnect between participants' reported GenAI usage and their perception of whether or not they have formed a habit of using the technology poses an interesting question not anticipated at the start of this project: are users aware of how much they use GenAI? Secondly, based on the content analysis of use cases, are they aware of how well GenAI performed on the tasks they prompted? Or are both of these perceptions subject to distortion? A usage tracker could track time spent by users interacting with the GenAI system, time spent reading outputs, and the amount of output copied out of the system, presumably to be pasted elsewhere. These metrics could help give users real information about how and how much they are using GenAI systems. Having such metrics may alter their perceptions of whether or not they have formed a habit of using GenAI.

A New Prompting Paradigm

While the above add-ons would be useful to any system they would rely on users opting into their use. There is a distinct possibility that users would simply ignore any additive information they are provided. Ideally, a system should fully integrate all aspects of system transparency and provide a dedicated support structure that actively teaches GenAI literacy concepts through use.

Along with the competencies, Long and Magerko (2020) also proposed a number of design considerations, including “Opportunities to Program” which they defined as ways for the user to teach the AI agent. While they focus on actually altering the code of the model, I would argue this consideration could also apply to the prompts given and the memories stored by the model during the execution of a given project. Currently, most consumer models have an interface similar to the one pictured below. ChatGPT provides, as of writing, a relatively unstructured experience. There is the dedicated Codex app for programming, the ability to store memories and upload sources for a specific project, and options to choose a general category of task on the front page, but there is little structure provided to users about how and when to best use

these features. The user is still largely expected to engineer an effective prompt on their own with little guidance.

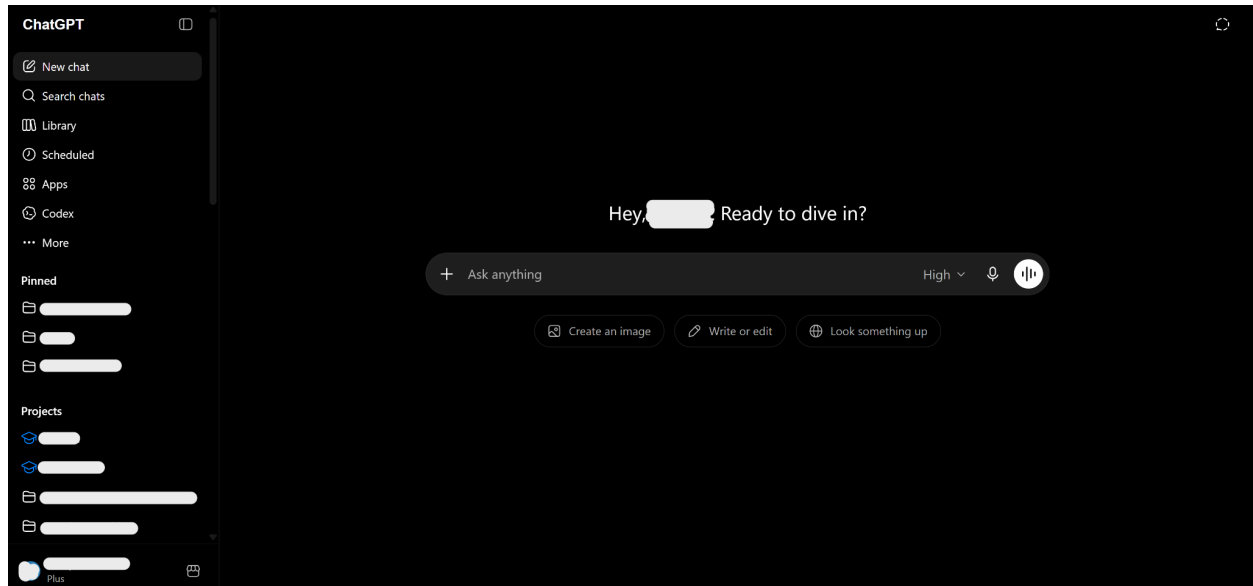


Fig 6. Screenshot of ChatGPT Pro subscription homepage as of 30.06.2026

The features already included in ChatGPT as an example, already lend themselves to being used to teach the model the project needs and obtain better output. However, the relatively unstructured interface, dominated by the prompt entry field, lends itself to users opening the site and firing off a quick prompt without necessarily thinking critically.

A structured walkthrough which gradually unveils all the possible methods of interacting with the model could help teach GenAI competencies and improve literacy while enabling users to get higher-quality outputs that they can evaluate better (Long & Magerko, 2020). Instead of opening directly to a free prompt field, model user interfaces could open with questions with selectable answer options about what task the user would like to perform. This would trigger a Socratic questioning method from the chatbot, wherein it would ask questions about the scope of the project, information needs, etcetera. By asking these questions the model could guide users to useful features and help users to address tasks in smaller, iterative chunks rather than one large prompt that may fail to capture the nuance of the project. A study by Nasr et al. (2025) found that this sort of iterative questioning, where users began by asking questions about the broad scope and then narrowing to each component, helped them break the complex task into subtasks and “analyze what information was needed to frame each query”. Users interviewed in that study also indicated that they found specificity and follow up questioning helped generate more useful responses (Nasr et al., 2025). This sort of questioning could also be used in a phased conversation to encourage properly going over the output. The chatbot could ask users to check its work before progressing. A user interface which promotes iterative questioning and guides users to useful features could drastically change how users interact with consumer GenAI models.

Conclusion

As GenAI becomes ubiquitous, it is important to consider how and how well users are putting the technology to work. Especially in a small business context, unchecked mistakes by recklessly using the output of GenAI could be disastrous. It begs the question, how can GenAI literacy be promoted among users who may or may not have a background in the underlying technology? Analysis of participants' GenAI literacy revealed troubling gaps in knowledge, and assessment of acceptance and content analysis revealed that participants may not have a complete grasp of how much time they spend using GenAI or how much caution to use when interpreting results. User interfaces with more informative add-ons and more structured conversations that helped guide an iterative prompting style could help to foster increased understanding of the strengths and weaknesses of GenAI as well as more effective and safer use of the technology.

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